21
Gröbner bases

In this chapter, we present an important algorithmic approach to dealing with polynomials in several variables. Hironaka (1964) introduced in his work on resolution of singularities over $\mathbb{C}$—for which he received the Fields medal, the "Nobel prize" in mathematics—a special type of basis for polynomial ideals, called "standard basis". Bruno Buchberger (1965) invented them independently in his Ph. D. thesis, and named them **Gröbner bases** after his thesis advisor Wolfgang Gröbner. They are a vital tool in modern computational algebraic geometry.

We start with two examples, one from robotics and one illustrating "automatic" proofs of theorems in geometry. We then introduce the basic notions of orders on monomials and the resulting division algorithm. Next come two important theorems, by Dickson and by Hilbert, that guarantee finite bases for certain ideals. Then we can define Gröbner bases and Buchberger's algorithm to compute them.

The end of this chapter presents two "geometric" applications: implicitization of algebraic varieties and solution of systems of polynomial equations. While these fall naturally into the realm of manipulating polynomials, the examples in Sections 24.1 and 24.2 below are less expected: logical proof systems and analysis of parallel processes. We cannot even mention numerous other applications, for example, in tiling problems and term rewriting. We finish with some facts—without proof—on the cost of computing Gröbner bases.

21.1. Polynomial ideals

We begin with two examples that illustrate the kind of problems that can be solved with the help of Gröbner bases.

EXAMPLE 21.1. Figure 21.1 shows a very simple robot, one-armed with two joints. The arm is fixed at one end with a joint to a point (say, the origin of the Cartesian plane), and has another joint in the middle. The distance between the two joints is 2, the joint between the two arms is in position (x, y), and the distance from the second joint to the endpoint, at position (z, w), is 1. Furthermore, there is

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

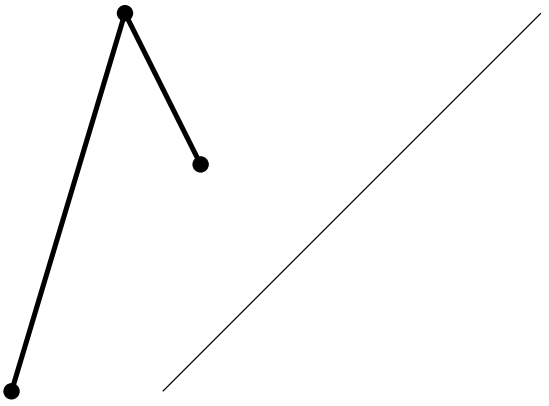


FIGURE 21.1: A two-segment robot.

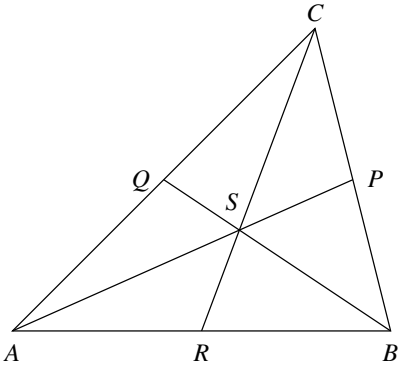


FIGURE 21.2: The three medians AP, BQ, and CR of a triangle ABC intersect at the center of gravity S.

a line $L = \{(u, \lambda u + \mu) : u \in \mathbb{R}\}$ with some fixed parameters $\lambda, \mu \in \mathbb{R}$. A simple question is: can the robot reach the line? The possible positions $(x, y, z, w) \in \mathbb{R}^4$ of the robot are characterized by the algebraic equations

$$x^2 + y^2 = 4 \text{ and } (z-x)^2 + (w-y)^2 = 1, \tag{1}$$

and an answer to the question is either a quadruple (x, y, z, w) satisfying (1) and the additional equation $w = \lambda z + \mu$, or a proof that no such quadruple exists. $\diamond$

EXAMPLE 21.2. A well-known geometric theorem says that the three medians of a triangle intersect at one point, the center of gravity of the triangle, and that the intersection point trisects each median (Figure 21.2). We now formulate this as

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

a problem about multivariate polynomials. Since the assumptions and the conclusion of the theorem are invariant under translation, rotation, and scaling, we may assume that two of the vertices of the triangle are $A = (0,0)$ and $B = (1,0)$, and the third point is $C = (x,y)$, with arbitrary $x,y \in \mathbb{R}$. Then the midpoints of the three edges BC, AC, and AB are $P = ((x+1)/2, y/2)$, $Q = (x/2, y/2)$, and $R = (1/2, 0)$, respectively. We let $S = (u,v)$ be the intersection point of the two medians AP and BQ. (If $y = 0$, then these two lines coincide.) The condition that S lies on AP is equivalent to saying that AS and AP have the same slope, so that

$$\frac{u}{v} = \frac{x+1}{y},$$

or, after clearing denominators,

$$f_1 = uy - v(x+1) = 0.$$

Similarly, the condition that S lies on BQ can be expressed as

$$f_2 = (u-1)y - v(x-2) = 0.$$

The claims now are that S also lies on the third median CR, or

$$g_1 = -2(u-x)y - (v-y)(1-2x) = -2uy - (v-y) + 2vx = 0,$$

and that S trisects each of the three medians, so that

$$\begin{aligned}(u,v) = AS = 2SP &= (x+1-2u, y-2v),\\ (u-1,v) = BS = 2SQ &= (x-2u, y-2v),\\ (u-x,v-y) = CS = 2SR &= (-2u+1, -2v),\end{aligned}$$

or equivalently,

$$g_2 = 3u - x - 1 = 0 \text{ and } g_3 = 3v - y = 0.$$

A short computation shows that $g_1 = -f_1 - f_2$, so that $g_1 = 0$ follows from $f_1 = f_2 = 0$, which establishes that the three medians intersect indeed in S. We will continue this example in Section 21.6. $\diamond$

Let F be a field, $R = F[x_1, \ldots, x_n]$ a polynomial ring in n variables over F, and $f_1, \ldots, f_s \in R$. The polynomials $f_1, \ldots, f_s$ **generate** (or form a **basis** of) the ideal

$$I = \langle f_1, \ldots, f_s \rangle = \left\{ \sum_{1 \le i \le s} q_i f_i : q_i \in R \right\}$$

(not to be confused with the *sequence* $(f_1, \ldots, f_s)$), and

$$V(I) = \{u \in F^n : f(u) = 0 \text{ for all } f \in I\} = \{u \in F^n : f_1(u) = \cdots = f_s(u) = 0\}$$

is the **variety of** I. We also write $V(f_1, \ldots, f_s)$ instead of $V(\langle f_1, \ldots, f_s \rangle)$ for short. Interesting questions about I include:

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

- Is $V(I) \neq \emptyset$?
- How "big" is $V(I)$?
- Ideal membership problem: given $f \in R$, is $f \in I$?
- Triviality: Is $I = R$?

EXAMPLE 21.2 (continued). In Example 21.2, we have seen that

$$g_1 = -f_1 - f_2 \in \langle f_1, f_2 \rangle \subseteq \mathbb{R}[u,v,x,y],$$

and used this to prove that $V(f_1, f_2) \subseteq V(f_3)$. If we can show that also $g_2, g_3 \in V(f_1, f_2)$, then this concludes the proof of the theorem. $\diamond$

EXAMPLE 21.3. (i) Let $f_1 = x^2 + y^2 - 1$ and $f_2 = y - 2$ in $R = \mathbb{R}[x,y]$, and $I = \langle f_1, f_2 \rangle$. Then

$$V(I) = \{(u,v) \in \mathbb{R}^2 : u^2 + v^2 - 1 = v - 2 = 0\} = \{(u,2) \in \mathbb{R}^2 : u^2 = -3\} = \emptyset$$

is the intersection of the circle $V(x^2 + y^2 - 1)$ with the line $V(y-2)$ (see Figure 21.3), which is empty over $\mathbb{R}$. If we regard f_1, f_2 as polynomials in $\mathbb{C}[x,y]$ and consider their variety over the complex numbers, then

$$V(I) = \{(u,2) \in \mathbb{C}^2 : u^2 = -3\} = \{(\sqrt{3}i, 2),\ (-\sqrt{3}i, 2)\}$$

consists of two points, where $i = \sqrt{-1} \in \mathbb{C}$.

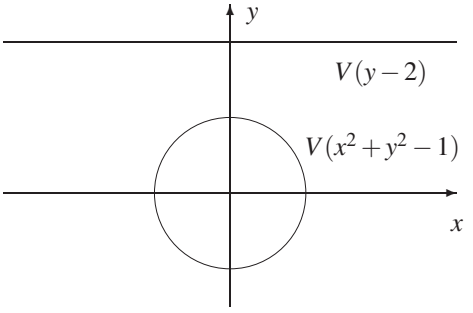


FIGURE 21.3: A circle and a line in $\mathbb{R}^2$.

(ii) Let $f = (y^2 + 6)(x-1) - y(x^2+1)$, $g = (x^2+6)(y-1) - x(y^2+1)$, $h = (x - 5/2)^2 + (y - 5/2)^2 - 1/2$ in $\mathbb{C}[x,y]$, and $I = \langle f, g \rangle$. We have seen in Example 6.41 that $V(I)$, the intersection of the two plane curves $V(f)$ and $V(g)$, consists of the six points

$$V(I) = \left\{ (2,2), (2,3), (3,2), (3,3), \left(\frac{1 \pm \sqrt{15}i}{2}, \frac{1 \mp \sqrt{15}i}{2} \right) \right\} \subseteq \mathbb{C}^2. \qquad (2)$$

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

A short calculation shows that $h = -f - g \in I$, and hence h vanishes at all points of $V(I)$ as well. Figure 6.3 shows the three curves $V(f)$, $V(g)$, $V(h)$, and their four real intersection points.

Now let $h^* = x^2 + y^2 - 5x - 5y + 11 \in \mathbb{C}[x,y]$. Then $V(h^*) \cap \mathbb{R}$ is a circle centered at $(5/2, 5/2)$ with a bigger radius than that of $V(h)$, and it contains none of the points in $V(I)$. Thus h^* does not belong to I. In fact, we have $-f - g - h^* = 1$ in $\mathbb{C}[x,y]$, and hence $\langle f, g, h^* \rangle = \mathbb{C}[x,y]$ and $V(f,g,h^*) = \emptyset$. Indeed, any common root of f, g, h^* would also be a "root" of 1, and hence there are no such common roots. Hilbert's famous **Nullstellensatz** (see Section 21.7) states that such a certificate always exists: when J is an ideal with $V(J) = \emptyset$, then $1 \in J$. This is true over $\mathbb{C}$, and any algebraically closed field, but, as (i) shows, not over $\mathbb{R}$. $\diamond$

The structure of these varieties is the subject of *algebraic geometry*, a deep mathematical theory whose most recent success is the solution of the 350–year old Fermat problem (page 514). We give a few pointers to the recent exciting work in *computational algebraic geometry* at the end of this chapter. A **Gröbner basis** of I is a special "basis" for I, in which these questions are easy to answer. Everything is easy in the case $n = 1$: Since $F[x]$ is a Euclidean domain (Chapter 3), we have

$$\langle f_1, \ldots, f_s \rangle = \langle \gcd(f_1, \ldots, f_s) \rangle \tag{3}$$

(Exercise 21.3), so that we may assume $s = 1$. Now we let $f, g \in F[x]$ and divide f by g with remainder, yielding $q, r \in F[x]$ with $f = qg + r$ and $\deg r < \deg g$. Then

$$f \in \langle g \rangle \iff r = 0, \tag{4}$$

and $V(g) = \{u_1, \ldots, u_d\}$ if $x - u_1, \ldots, x - u_d$ are the distinct linear factors of g in $F[x]$. Equation (3) is no longer true in the case of two variables or more. For example, $\gcd(x,y) = 1$ in $F[x,y]$, but $\langle x, y \rangle$ is different from $\langle 1 \rangle = F[x,y]$ (Exercise 21.1). Much of the effort of Gröbner bases goes into restoring some of these apparently lost properties, such as division with remainder.

21.2. Monomial orders and multivariate division with remainder

We introduce an analog of division with remainder in the case of many variables. In order to do that, we have to say what the "leading term" of a polynomial should be in the general case.

A **partial order** $<$ on a set S is an irreflexive and transitive relation, so that

$$\text{not } (\alpha < \alpha) \text{ and } \alpha < \beta < \gamma \Longrightarrow \alpha < \gamma \text{ for all } \alpha, \beta, \gamma \in S.$$

These conditions imply that $<$ is asymmetric: $((\alpha < \beta) \text{ and } (\beta < \alpha))$ is always false (Exercise 21.7). A partial order is a **total order** (or simply order) if either

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

$\alpha = \beta$ or $\alpha < \beta$ or $\beta < \alpha$, for all $\alpha, \beta \in S$, and is a **well-order** if in addition every nonempty subset of S has a least element. If $<$ is a partial order, we write $\alpha \le \beta$ if $\alpha < \beta$ or $\alpha = \beta$, and $\alpha > \beta$ or $\alpha \ge \beta$ if $\beta < \alpha$ or $\beta \le \alpha$, respectively. For example, the familiar orders $<$ on $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are total, but only the first one is a well-order. The inclusion $\subseteq$ on the set $2^{\mathbb{N}}$ of subsets of $\mathbb{N}$ is a partial order, but not total.

We identify the vector $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ with the monomial

$$\boldsymbol{x}^{\alpha} = x_1^{\alpha_1} \cdots x_n^{\alpha_n} \in R.$$

DEFINITION 21.4. *A **monomial order** in $R = F[x_1, \dots, x_n]$ is a relation $\prec$ on $\mathbb{N}^n$ such that*

(i) $\prec$ *is a total order,*

(ii) $\alpha \prec \beta \Longrightarrow \alpha + \gamma \prec \beta + \gamma$ *for all* $\alpha, \beta, \gamma \in \mathbb{N}^n$,

(iii) $\prec$ *is a well-order.*

Under the above identification, this gives an order on the monomials in R. If $n = 1$, for example, the natural order on $\mathbb{N} = \mathbb{N}^1$ is a monomial order, and the corresponding univariate monomials are ordered by their degree.

If (i) holds, then (iii) is equivalent to the condition that there be no infinite descending chain $\alpha_1 \succ \alpha_2 \succ \alpha_3 \succ \cdots$ in $\mathbb{N}^n$.

EXAMPLE 21.5. Here are three standard examples of monomial orders:

(i) **Lexicographic order**:

$$\alpha \prec_{\text{lex}} \beta \iff \text{the leftmost nonzero entry in } \alpha - \beta \text{ is negative.}$$

For example, if $n = 3$, $\alpha_1 = (0,4,0)$, $\alpha_2 = (1,1,2)$, $\alpha_3 = (1,2,1)$, and $\alpha_4 = (3,0,0)$, then $\alpha_1 \prec_{\text{lex}} \alpha_2 \prec_{\text{lex}} \alpha_3 \prec_{\text{lex}} \alpha_4$.

(ii) **Graded lexicographic order**: Let $\alpha = (\alpha_1, \dots, \alpha_n), \beta = (\beta_1, \dots, \beta_n) \in \mathbb{N}^n$. Then

$$\alpha \prec_{\text{grlex}} \beta \iff \sum_{1 \le i \le n} \alpha_i < \sum_{1 \le i \le n} \beta_i \text{ or } \Big(\sum_{1 \le i \le n} \alpha_i = \sum_{1 \le i \le n} \beta_i \text{ and } \alpha \prec_{\text{lex}} \beta \Big).$$

With $\alpha_1, \dots, \alpha_4$ as above, we have $\alpha_4 \prec_{\text{grlex}} \alpha_1 \prec_{\text{grlex}} \alpha_2 \prec_{\text{grlex}} \alpha_3$.

(iii) **Graded reverse lexicographic order**:

$$\alpha \prec_{\text{grevlex}} \beta \iff \sum_{1 \le i \le n} \alpha_i < \sum_{1 \le i \le n} \beta_i \text{ or } \Big(\sum_{1 \le i \le n} \alpha_i = \sum_{1 \le i \le n} \beta_i \text{ and the rightmost nonzero entry in } \alpha - \beta \in \mathbb{Z}^n \text{ is positive} \Big).$$

Then $\alpha_4 \prec_{\text{grevlex}} \alpha_2 \prec_{\text{grevlex}} \alpha_3 \prec_{\text{grevlex}} \alpha_1$, with $\alpha_1, \dots, \alpha_4$ as before. $\diamond$

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

In all three examples, the variables (that is, the monomials of degree one) are ordered as $x_1 \succ x_2 \succ \ldots \succ x_{n-1} \succ x_n$. "Graded" refers to the fact that the total degree $\sum \alpha_i$ is the main criterion. In the case $n = 1$, we have $\prec_{\text{lex}} = \prec_{\text{grlex}} = \prec_{\text{grevlex}}$.

Once we have a monomial order on R, we can sort terms of a polynomial according to $\prec$.

EXAMPLE 21.5 (continued). Let $f = 4xyz^2 + 4x^3 - 5y^4 + 7xy^2z \in \mathbb{Q}[x,y,z]$ (we always identify x,y,z with x_1,x_2,x_3). Then the orders of f with respect to $\prec_{\text{lex}}$, $\prec_{\text{grlex}}$, and $\prec_{\text{grevlex}}$ are: $4x^3 + 7xy^2z + 4xyz^2 - 5y^4$, $7xy^2z + 4xyz^2 - 5y^4 + 4x^3$, and $-5y^4 + 7xy^2z + 4xyz^2 + 4x^3$, respectively. $\diamond$

THEOREM 21.6.

$\prec_{\text{lex}}$, $\prec_{\text{grlex}}$, and $\prec_{\text{grevlex}}$, are monomial orders.

PROOF. The proof is a simple check; we give some details only for $\prec_{\text{grevlex}}$. We omit the verification that $\prec_{\text{grevlex}}$ is a partial order. For each $\alpha, \beta \in \mathbb{N}^n$ with $\alpha \neq \beta$, we have either $\sum_{1\le i\le n} \alpha_i < \sum_{1\le i\le n} \beta_i$, or $\sum_{1\le i\le n} \beta_i < \sum_{1\le i\le n} \alpha_i$, or $\sum_{1\le i\le n} \alpha_i = \sum_{1\le i\le n} \beta_i$, and in the last case either the rightmost nonzero entry in $\alpha - \beta$ is positive or the rightmost nonzero entry in $\beta - \alpha$ is positive. Thus $\prec_{\text{grevlex}}$ is total.

For condition (ii), we have

$$\sum_{1\le i\le n} \alpha_i < \sum \beta_i \iff \sum_{1\le i\le n} (\alpha_i + \gamma_i) < \sum_i (\beta_i + \gamma_i),$$

and similarly for "=", and $\alpha - \beta = (\alpha + \gamma) - (\beta + \gamma)$. For condition (iii), if $S \subseteq \mathbb{N}^n$ is nonempty and $T \subseteq S$ the set of monomials of smallest total degree in S, then T is finite (since for any $m \in \mathbb{N}$ there are only finitely many monomials of total degree m) and $\min T = \min S$. $\square$

The antilexicographic order $\prec_{\text{alex}}$ on $\mathbb{N}^2$, with $\alpha \prec_{\text{alex}} \beta \iff \beta \prec_{\text{lex}} \alpha$, is an example where condition (iii) is violated: $S = \mathbb{N} \times \{0\}$ has no smallest element, since $(0,0) \succ_{\text{alex}} (1,0) \succ_{\text{alex}} (2,0) \succ_{\text{alex}} \cdots$.

DEFINITION 21.7. *Let $f = \sum_{\alpha\in\mathbb{N}^n} c_\alpha \boldsymbol{x}^\alpha \in R$ be a nonzero polynomial with all $c_\alpha \in F$ (only finitely many nonzero), and $\prec$ a monomial order.*

(i) Each $c_\alpha \boldsymbol{x}^\alpha$ with $c_\alpha \neq 0$ is a ***term*** *of f.*

(ii) The ***multidegree*** *of f is $\operatorname{mdeg}(f) = \max_\prec\{\alpha \in \mathbb{N}^n : c_\alpha \neq 0\}$, where $\max_\prec$ is the maximum with respect to $\prec$.*

(iii) The ***leading coefficient*** *of f is $\operatorname{lc}(f) = c_{\operatorname{mdeg}(f)} \in F \setminus \{0\}$.*

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

(iv) *The* ***leading monomial*** *of* f *is* $\operatorname{lm}(f) = \boldsymbol{x}^{\operatorname{mdeg}(f)} \in R$.

(v) *The* ***leading term*** *of* f *is* $\operatorname{lt}(f) = \operatorname{lc}(f) \cdot \operatorname{lm}(f) \in R$.

EXAMPLE 21.5 (continued). To illustrate these notions, we take the polynomial $f = 4xyz^2 + 4x^3 - 5y^4 + 7xy^2z \in \mathbb{Q}[x,y,z]$ and the three orders from Example 21.5.

	$\prec_{\text{lex}}$	$\prec_{\text{grlex}}$	$\prec_{\text{grevlex}}$
$\operatorname{mdeg}(f)$	$(3,0,0)$	$(1,2,1)$	$(0,4,0)$
$\operatorname{lc}(f)$	4	7	-5
$\operatorname{lm}(f)$	x^3	xy^2z	y^4
$\operatorname{lt}(f)$	$4x^3$	$7xy^2z$	$-5y^4$

$\diamond$

If we wanted to extend these definitions to the zero polynomial, we would formally adjoin the element $-\infty$ to $\mathbb{N}^n$ and let $\operatorname{lt}(0) = \operatorname{lm}(0) = \operatorname{lc}(0) = 0$ and $\operatorname{mdeg}(0) = -\infty$, with the obvious interpretation. Also, $\alpha \preccurlyeq \beta$ means that $\alpha \prec \beta$ or $\alpha = \beta$.

The following lemma is proved in Exercise 21.11.

LEMMA 21.8. *Let* $\prec$ *be a monomial order on* R*, and* $f, g \in R \setminus \{0\}$.

(i) $\operatorname{mdeg}(fg) = \operatorname{mdeg}(f) + \operatorname{mdeg}(g)$.

(ii) *If* $f + g \neq 0$ *then* $\operatorname{mdeg}(f+g) \preccurlyeq \max\{\operatorname{mdeg}(f), \operatorname{mdeg}(g)\}$, *with equality if* $\operatorname{mdeg}(f) \neq \operatorname{mdeg}(g)$.

Our next goal is an algorithm for division with remainder in R. Given polynomials $f, f_1, \ldots, f_s \in R$, we want to write $f = q_1 f_1 + \ldots + q_s f_s + r$ with $q_1, \ldots, q_s, r$ in R. Before stating the algorithm formally, we give some examples.

EXAMPLE 21.9. Let $\prec = \prec_{\text{lex}}$, $f = xy^2 + 1$, $f_1 = xy + 1$, $f_2 = y + 1$.

	$xy+1$	$y+1$
xy^2+1	y	
$-(xy^2+y)$		
$-y+1$		-1
$-(-y-1)$		
2		

	$xy+1$	$y+1$
xy^2+1		xy
$-(xy^2+xy)$		
$-xy+1$		$-x$
$-(-xy-x)$		
$x+1$		

In the left hand table, division is performed as in the univariate case, with the difference that we have two divisors instead of one. The quotient of the two leading terms that we get in each step is recorded in the column below the respective divisor. In the last line, 2 is not divisible by the leading term of f_1 or f_2,

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

and the process stops. Hence $f = y \cdot f_1 - 1 \cdot f_2 + 2$. Note that there is some non-determinism here because we might as well have chosen f_2 instead of f_1 in the first step; this is illustrated in the right hand table, where we find that $f = 0 \cdot f_1 + (xy - x) \cdot f_2 + (x + 1)$, and no term in $x + 1$ is divisible by $\mathrm{lc}(f_1)$ or $\mathrm{lc}(f_2)$. $\diamond$

EXAMPLE 21.10. Let $\prec = \prec_{\mathrm{lex}}$, $f = x^2y + xy^2 + y^2$, $f_1 = xy - 1$, $f_2 = y^2 - 1$.

	$xy-1$	y^2-1	remainder
$x^2y+xy^2+y^2$	x		
$-(x^2y-x)$			
xy^2+x+y^2	y		
$-(xy^2-y)$			
$x+y^2+y$			x
$-x$			
y^2+y		1	
$-(y^2-1)$			
$y+1$			

Here, a phenomenon occurs that cannot happen in the univariate case. In the third step, the leading term x is not divisible by the leading term of f_1 or f_2 and is moved to the remainder column. Afterwards, the division may continue further, and the result is $f = (x + y) \cdot f_1 + 1 \cdot f_2 + (x + y + 1)$. $\diamond$

ALGORITHM 21.11 Multivariate division with remainder.

Input: Nonzero polynomials $f, f_1, \ldots, f_s \in R = F[x_1, \ldots, x_n]$, where F is a field, and a monomial order $\prec$ on R.

Output: $q_1, \ldots, q_s, r \in R$ such that $f = q_1 f_1 + \cdots + q_s f_s + r$ and no monomial in r is divisible by any of $\mathrm{lt}(f_1), \ldots, \mathrm{lt}(f_s)$.

1. $r \longleftarrow 0, \quad p \longleftarrow f$
   for $i = 1, \ldots, s$ **do** $q_i \longleftarrow 0$
2. **while** $p \neq 0$ **do**
3. **if** $\mathrm{lt}(f_i)$ divides $\mathrm{lt}(p)$ for some $i \in \{1, \ldots, s\}$
   then choose some such i, $\quad q_i \longleftarrow q_i + \dfrac{\mathrm{lt}(p)}{\mathrm{lt}(f_i)}, \quad p \longleftarrow p - \dfrac{\mathrm{lt}(p)}{\mathrm{lt}(f_i)} f_i$
   else $r \longleftarrow r + \mathrm{lt}(p), \quad p \longleftarrow p - \mathrm{lt}(p)$
4. **return** $q_1, \ldots, q_s, r$.

The following theorem implies the correctness of the algorithm; its proof is left as Exercise 21.12.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

THEOREM 21.12.
Each time the algorithm passes through step 3, the following invariants hold.

(i) $\operatorname{mdeg}(p) \preccurlyeq \operatorname{mdeg}(f)$ *and* $f = p + q_1 f_1 + \cdots + q_s f_s + r$,

(ii) $q_i \neq 0 \Longrightarrow \operatorname{mdeg}(q_i f_i) \preccurlyeq \operatorname{mdeg}(f)$ *for* $1 \leq i \leq s$,

(iii) no term in r *is divisible by any* $\operatorname{lt}(f_i)$.

This kind of division with remainder need not be unique: there may be a choice for the value of i in step 3 when the leading term of f is divisible by more than one $\operatorname{lt}(f_i)$. We have already encountered this in Example 21.9, and here is another one.

EXAMPLE 21.10 (continued). If we choose f_2 instead of f_1 in the second step of Example 21.10, we get

	$xy-1$	y^2-1	remainder
$x^2y+xy^2+y^2$	x		
$-(x^2y-x)$			
xy^2+x+y^2		x	
$-(xy^2-x)$			
$2x+y^2$			$2x$
$-2x$			
y^2		1	
$-(y^2-1)$			
1			

and $f = x \cdot f_1 + (x+1) \cdot f_2 + (2x+1)$. $\diamond$

If we make Algorithm 21.11 deterministic by always choosing the smallest possible i in step 3, then the **quotients** $q_1, \ldots, q_s$, and the **remainder** r, for which we write

$$r = f \text{ rem } (f_1, \ldots, f_s), \tag{5}$$

are uniquely determined. For $s = 1$, division with remainder solves the ideal membership problem: $f \in \langle f_1 \rangle$ holds if and only if the remainder is zero. This fails in general for $s \geq 2$, as the following example shows.

EXAMPLE 21.13. We divide $f = xy^2 - x$ by $f_1 = xy + 1$ and $f_2 = y^2 - 1$:

	$xy+1$	y^2-1
xy^2-x	y	
$-(xy^2+y)$		
$-x-y$		

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

so that $f = y \cdot f_1 + 0 \cdot f_2 + (-x-y)$. But on the other hand, $f = 0 \cdot f_1 + x \cdot f_2 + 0$, which shows that $f \in \langle f_1, f_2 \rangle$. $\diamond$

Our goal is now to find a special basis of an arbitrary ideal such that the remainder on division by that basis is unique and thus gives the correct answer to the ideal membership problem, as in (4) for $n = 1$. At first sight, it is not clear whether such a type of basis exists at all.

21.3. Monomial ideals and Hilbert's basis theorem

DEFINITION 21.14. *A **monomial ideal** $I \subseteq R$ is an ideal generated by monomials in R, so that there exists a subset $A \subseteq \mathbb{N}^n$ with*

$$I = \langle \boldsymbol{x}^A \rangle = \langle \{\boldsymbol{x}^\alpha : \alpha \in A\} \rangle.$$

LEMMA 21.15. *Let $I = \langle \boldsymbol{x}^A \rangle \subseteq R$ be a monomial ideal, and $\beta \in \mathbb{N}^n$. Then*

$$\boldsymbol{x}^\beta \in I \quad \Longleftrightarrow \quad \exists \alpha \in A \quad \boldsymbol{x}^\alpha \mid \boldsymbol{x}^\beta.$$

PROOF. "$\Longleftarrow$" is clear. Conversely, let $\alpha_1, \ldots, \alpha_s \in A$ and $q_1, \ldots, q_s \in R$ such that $\boldsymbol{x}^\beta = \sum_i q_i \boldsymbol{x}^{\alpha_i}$. There is at least one term $q_i \boldsymbol{x}^{\alpha_i}$ in which $\boldsymbol{x}^\beta$ occurs, and then $\boldsymbol{x}^{\alpha_i} \mid \boldsymbol{x}^\beta$. $\square$

LEMMA 21.16. *Let $I \subseteq R$ be a monomial ideal and $f \in R$. Then the following are equivalent:*

(i) $f \in I$,

(ii) *each term of f is in I,*

(iii) *f is an F–linear combination of monomials in I.*

PROOF. (i) $\Longrightarrow$ (ii): As in the proof of Lemma 21.15, a representation $f = \sum_i q_i \boldsymbol{x}^{\alpha_i}$ implies that each monomial of f is divisible by some $\boldsymbol{x}^\gamma$, with $\gamma \in A$. The implications (ii) $\Longrightarrow$ (iii) $\Longrightarrow$ (i) are clear, and, in fact, hold for any ideal I. $\square$

For example, if $I = \langle x^3, x^2 y \rangle \subseteq \mathbb{Q}[x,y]$, then the lemma shows that $3x^4 + 5x^2y^3 \in I$ and $2x^4 y + 7x^2 \notin I$. The implication (i) $\Longrightarrow$ (ii) is false for some ideals, as shown in Example 21.21 below.

COROLLARY 21.17.
Two monomial ideals are identical if and only if they contain the same monomials.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

THEOREM 21.18 *Dickson's lemma.*
Every monomial ideal is generated by a finite set of monomials. More precisely, for every $A \subseteq \mathbb{N}^n$ there exists a finite subset $B \subseteq A$ such that $\langle \boldsymbol{x}^A \rangle = \langle \boldsymbol{x}^B \rangle$.

PROOF. Except for its last sentence, our proof is purely combinatorial, without any algebra. The claim is trivial if $A = \emptyset$, and we may assume that A is nonempty. We define a relation "$\leq$" on $\mathbb{N}^n$ by

$$\alpha \leq \beta \iff \alpha_i \leq \beta_i \text{ for } 1 \leq i \leq n,$$

for all $\alpha = (\alpha_1, \ldots, \alpha_n)$ and $\beta = (\beta_1, \ldots, \beta_n)$ in $\mathbb{N}^n$, so that $\alpha \leq \beta$ if and only if $\boldsymbol{x}^\alpha \mid \boldsymbol{x}^\beta$. As usual, we write $\alpha < \beta$ if $\alpha \leq \beta$ and $\alpha \neq \beta$. Then $<$ is a partial order on $\mathbb{N}^n$ which is not total if $n \geq 2$ (for example, we have neither $(1,0) < (0,1)$ nor $(1,0) > (0,1)$). We now let $B = \{\alpha \in A : \forall \beta \in A \ \beta \not< \alpha\}$ be the set of minimal elements of A with respect to $\leq$, and claim that B is a finite subset of A and

$$\text{for each } \alpha \in A \text{ there exists some } \beta \in B \text{ with } \beta \leq \alpha. \tag{6}$$

For any $\alpha \in \mathbb{N}^n$ there are only finitely many $\beta \in \mathbb{N}^n$ with $\beta \leq \alpha$ (Exercise 21.13), and hence there is no infinite descending chain of elements $\alpha^{(1)} > \alpha^{(2)} > \alpha^{(3)} > \cdots$ in $\mathbb{N}^n$. In particular, for any $\alpha \in A$ there is some minimal element $\beta \in B$ such that $\beta \leq \alpha$.

It remains to show that B is finite, which we prove by induction on n. If $n = 1$, then $<$ is a total order, and B consists of the unique smallest element of A. If $n \geq 2$, we let $A^* = \{(\alpha_1, \ldots, \alpha_{n-1}) \in \mathbb{N}^{n-1} : \exists \alpha_n \in \mathbb{N} \ (\alpha_1, \ldots, \alpha_n) \in A\}$. By the induction hypothesis, the set B^* of minimal elements of A^* is finite. For each $\beta = (\beta_1, \ldots, \beta_{n-1}) \in B^*$, we choose some $b_\beta \in \mathbb{N}$ such that $(\beta_1, \ldots, \beta_{n-1}, b_\beta) \in A$, and let $b = \max\{b_\beta : \beta \in B^*\}$. We claim that every $(\alpha_1, \ldots, \alpha_n) \in B$ has $\alpha_n \leq b$. Let $\alpha = (\alpha_1, \ldots, \alpha_n) \in B$. Then there exists some minimal element $\beta = (\beta_1, \ldots, \beta_{n-1}) \in B^*$ of A^* such that $\beta \leq (\alpha_1, \ldots, \alpha_{n-1})$. If $\alpha_n > b$, then

$$(\beta_1, \ldots, \beta_{n-1}, b_\beta) \leq (\beta_1, \ldots, \beta_{n-1}, b) < \alpha,$$

and α is not minimal. This proves the claim, and similarly we also find that all other coordinates of minimal elements are bounded, which implies that there are only finitely many of them.

Now (6) and the fact that $\alpha \leq \beta \iff \boldsymbol{x}^\alpha \mid \boldsymbol{x}^\beta$ imply that $\boldsymbol{x}^A \subseteq \langle \boldsymbol{x}^B \rangle$, whence $\langle \boldsymbol{x}^A \rangle \subseteq \langle \boldsymbol{x}^B \rangle$, and the reverse inclusion follows trivially from $B \subseteq A$. □

EXAMPLE 21.19. Let $n = 2$ and $A = \{(\alpha_1, \alpha_2) \in \mathbb{N}^2 : 6\alpha_2 = \alpha_1^2 - 7\alpha_1 + 18\}$. Then the set of minimal elements is $B = \{(0,3), (1,2), (3,1)\}$, as can be seen from Figure 21.4, and hence $\langle \boldsymbol{x}^A \rangle = \langle y^3, xy^2, x^3 y \rangle$. ◇

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

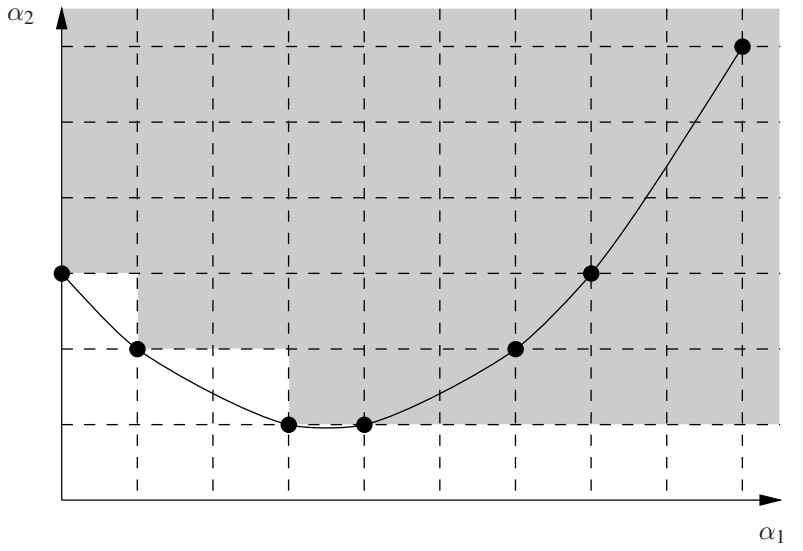


FIGURE 21.4: The monomial ideal $I = \langle \boldsymbol{x}^A \rangle$, where $A = \{(\alpha_1, \alpha_2) \in \mathbb{N}^2 : f(\alpha_1, \alpha_2) = 0\}$ and $f(\alpha_1, \alpha_2) = 6\alpha_2 - \alpha_1^2 + 7\alpha_1 - 18$. The elements of A are marked by a $\bullet$, the black curve is the set of all *real* (α_1, α_2) such that $f(\alpha_1, \alpha_2) = 0$, and a monomial $\boldsymbol{x}^\alpha$ is contained in I if and only if α lies in the gray area.

COROLLARY 21.20.
Let $\prec$ be a total order on $\mathbb{N}^n$ such that

$$\forall \alpha, \beta, \gamma \in \mathbb{N} \quad \alpha \prec \beta \Longrightarrow \alpha + \gamma \prec \beta + \gamma.$$

Then $\prec$ is a well-order if and only if $\alpha \succcurlyeq 0$ for all $\alpha \in \mathbb{N}^n$.

PROOF. We only prove "$\Longleftarrow$"; the reverse implication is Exercise 21.8. Let $A \subseteq \mathbb{N}^n$ be nonempty, and $I = \langle \boldsymbol{x}^A \rangle \subseteq R$. Then I is finitely generated, by Dickson's lemma, and

$$\exists \alpha_1, \ldots, \alpha_s \in A \quad I = \langle \boldsymbol{x}^{\alpha_1}, \ldots, \boldsymbol{x}^{\alpha_s} \rangle.$$

We order them so that $\alpha_1 \prec \alpha_2 \prec \cdots \prec \alpha_s$, and claim that $\min_\prec A = \alpha_1$. Let $\alpha \in A$ be arbitrary. Since $\boldsymbol{x}^\alpha \in I$, by Lemma 21.15 there exist $i \leq s$ and $\gamma \in \mathbb{N}^n$ with $\alpha = \alpha_i + \gamma$. Thus $\alpha = \alpha_i + \gamma \succcurlyeq \alpha_1 + \gamma \succcurlyeq \alpha_1 + 0 = \alpha_1$, and hence $\alpha_1 = \min_\prec A$. □

Thus we can replace the condition (iii) in the definition of monomial orders by

(iii)' $\quad \forall \alpha \in \mathbb{N}^n \quad \alpha \succcurlyeq 0.$

For any subset $G \subseteq R$ different from $\emptyset$ and $\{0\}$, we let $\mathrm{lt}(G) = \{\mathrm{lt}(g) : g \in G\}$. If $I \subseteq R$ is an ideal, then there is a finite subset $G \subseteq I$ such that $\langle \mathrm{lt}(G) \rangle = \langle \mathrm{lt}(I) \rangle$, by Dickson's lemma. However, it can happen that a finite set G generates I but $\langle \mathrm{lt}(G) \rangle \subsetneq \langle \mathrm{lt}(I) \rangle$, as in the following example.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

EXAMPLE 21.21. Let $g = x^3 - 2xy, h = x^2y - 2y^2 + x \in \mathbb{Q}[x,y]$, $\prec = \prec_{\text{grlex}}$, $G = \{g,h\}$, and $I = \langle G \rangle$. Then $x^2 = -y \cdot g + x \cdot h$ and $x^2 \in \langle \text{lt}(I) \rangle$ but $x^2 \notin \langle \text{lt}(G) \rangle = \langle x^3, x^2y \rangle$. $\diamond$

LEMMA 21.22. *Let I be an ideal in $R = F[x_1, \ldots, x_n]$. If $G \subseteq I$ is a finite subset such that $\langle \text{lt}(G) \rangle = \langle \text{lt}(I) \rangle$, then $\langle G \rangle = I$.*

PROOF. Let $G = \{g_1, \ldots, g_s\}$. If f is an arbitrary polynomial in I, then division with remainder yields $f = q_1 g_1 + \cdots + q_s g_s + r$, with $q_1, \ldots, q_s, r \in R$, such that either $r = 0$ or no term of r is divisible by the leading term of any g_i. But $r = f - q_1 g_1 - \cdots - q_s g_s \in I$, and hence $\text{lt}(r) \in \text{lt}(I) \subseteq \langle \text{lt}(g_1), \ldots, \text{lt}(g_s) \rangle$. This together with Lemma 21.15 implies that $r = 0$. Thus $f \in \langle g_1, \ldots, g_s \rangle = \langle G \rangle$. $\square$

Together with Dickson's lemma, applied to $\langle \text{lt}(I) \rangle$, and the fact that the zero ideal $\{0\}$ is generated by the zero polynomial, we obtain the following famous result.

THEOREM 21.23 *Hilbert's basis theorem.*
Every ideal I in $R = F[x_1, \ldots, x_n]$ is finitely generated. More precisely, there exists a finite subset $G \subseteq I$ such that $\langle G \rangle = I$ and $\langle \text{lt}(G) \rangle = \langle \text{lt}(I) \rangle$.

An immediate consequence is the following corollary.

COROLLARY 21.24 *Ascending chain condition.*
Let $I_1 \subseteq I_2 \subseteq I_3 \subseteq \ldots$ be an ascending chain of ideals in R. Then the chain stabilizes, so that $I_n = I_{n+1} = I_{n+2} = \cdots$ for some $n \in \mathbb{N}$.

PROOF. Let $I = \bigcup_{j \geq 1} I_j$. Then I is an ideal, which is finitely generated, by Hilbert's basis theorem, say $I = \langle g_1, \ldots, g_s \rangle$. With $n = \min\{j \geq 1 : g_1, \ldots, g_s \in I_j\}$, we then have $I_n = I_{n+1} = \cdots = I$. $\square$

In general, a ring satisfying the ascending chain condition is called **Noetherian**, after Emmy Noether. Thus $F[x_1, \ldots, x_n]$ is Noetherian if F is a field.

21.4. Gröbner bases and S-polynomials

Hilbert's basis theorem motivates the following definition.

DEFINITION 21.25. *Let $\prec$ be a monomial order and $I \subseteq R$ an ideal. A finite set $G \subseteq I$ is a **Gröbner basis** for I with respect to $\prec$ if $\langle \text{lt}(G) \rangle = \langle \text{lt}(I) \rangle$.*

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

Lemma 21.22 says that any Gröbner basis G for I is in fact a basis of I in the ring theoretic sense, which means that $\langle G\rangle = I$. With the convention that $\langle\rangle = \langle\emptyset\rangle = \{0\}$, Hilbert's basis theorem implies the following.

COROLLARY 21.26.
Every ideal I in $R = F[x_1,\dots,x_n]$ has a Gröbner basis.

In Example 21.21, G is not a Gröbner basis for I, but $\{g,h,x^2,2xy,-2y^2+x\}$ is, as we will see below.

We now want to show that division with remainder by a Gröbner basis is a valid ideal membership test. Throughout this section, we assume some monomial order on R.

LEMMA 21.27. *Let G be a Gröbner basis for an ideal $I \subseteq R$, and $f \in R$. Then there is a unique polynomial $r \in R$ with*

(i) $f - r \in I$,

(ii) no term of r is divisible by any monomial in $\mathrm{lt}(G)$.

PROOF. Existence follows from Theorem 21.12. For the uniqueness, we suppose that $f = h_1 + r_1 = h_2 + r_2$ with $h_1, h_2 \in I$ and no term of r_1 or r_2 divisible by any of $\mathrm{lt}(G)$. Then $r_1 - r_2 = h_2 - h_1 \in I$, and $\mathrm{lt}(r_1 - r_2)$ is divisible by some $\mathrm{lt}(g)$ with $g \in G$, by Lemma 21.15. Hence $r_1 - r_2 = 0$. □

Thus the remainder r on division of f by G does not depend on the order in which we divide by the elements of G. Extending the notation (5), we then write

$$f \text{ rem } G = r \in R.$$

The next result follows immediately from the lemma.

THEOREM 21.28.
Let G be a Gröbner basis for the ideal $I \subseteq R$ with respect to a monomial order $\prec$, and $f \in R$. Then $f \in I$ if and only if $f \text{ rem } G = 0$.

This property of G is equivalent to being a Gröbner basis (Exercise 21.17). It solves the ideal membership problem: all we have to do in order to test whether $f \in I$ is to divide f by G.

However, the proof of Hilbert's basis theorem is not constructive: it does not tell us how to compute a Gröbner basis for an ideal I from a given basis G. We now

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

solve this computational problem, so that we obtain an effective version of Hilbert's basis theorem. In order to construct a Gröbner basis, we investigate how G can fail to be a Gröbner basis. One possible reason is that a linear combination $a\boldsymbol{x}^{\alpha}g + b\boldsymbol{x}^{\beta}h$ of two polynomials $g,h \in G$ with $a,b \in F$ and $\alpha,\beta \in \mathbb{N}^n$ may yield a polynomial whose leading term is not divisible by any of $\mathrm{lt}(G)$, by cancellation of leading terms. In Example 21.21, we had $(-y)\cdot g + x\cdot h = x^2$, and $\mathrm{lt}(x^2) = x^2$ is not divisible by any of $\mathrm{lt}(G) = \{x^3, x^2y\}$.

DEFINITION 21.29. *Let $g,h \in R$ be nonzero, $\alpha = (\alpha_1,\ldots,\alpha_n) = \mathrm{mdeg}(g)$, $\beta = (\beta_1,\ldots,\beta_n) = \mathrm{mdeg}(h)$, and $\gamma = (\max\{\alpha_1,\beta_1\},\ldots,\max\{\alpha_n,\beta_n\})$. The* ***S-polynomial*** *of g and h is*

$$S(g,h) = \frac{\boldsymbol{x}^{\gamma}}{\mathrm{lt}(g)}g - \frac{\boldsymbol{x}^{\gamma}}{\mathrm{lt}(h)}h \in R. \tag{7}$$

Clearly $S(h,g) = -S(g,h)$, and since $\boldsymbol{x}^{\gamma}/\mathrm{lt}(g), \boldsymbol{x}^{\gamma}/\mathrm{lt}(h) \in R$, we have $S(g,h) \in \langle g,h\rangle$. In Example 21.21, we have $\alpha = (3,0)$, $\beta = (2,1)$, $\gamma = (3,1)$, and

$$S(g,h) = \frac{x^3y}{x^3}g - \frac{x^3y}{x^2y}h = -x^2.$$

The following lemma says that when cancellation of leading terms occurs in a linear combination of polynomials in G, it necessarily comes from S-polynomials.

LEMMA 21.30. *Let $g_1,\ldots,g_s \in R$, $\alpha_1,\ldots,\alpha_s \in \mathbb{N}^n$, $c_1,\ldots,c_s \in F \setminus \{0\}$,*

$$f = \sum_{1\le i\le s} c_i\boldsymbol{x}^{\alpha_i}g_i \in R, \tag{8}$$

and $\delta \in \mathbb{N}^n$ such that $\alpha_i + \mathrm{mdeg}(g_i) = \delta$ for $1 \le i \le s$, and $\mathrm{mdeg}(f) \prec \delta$. (This expresses the assumption that the leading terms cancel.) Then $\boldsymbol{x}^{\gamma_{ij}}$ divides $\boldsymbol{x}^{\delta}$ for $1 \le i < j \le s$, where $\boldsymbol{x}^{\gamma_{ij}} = \mathrm{lcm}(\mathrm{lm}(g_i),\mathrm{lm}(g_j))$, and there are $c_{ij} \in F$ such that

$$f = \sum_{1\le i<j\le s} c_{ij}\boldsymbol{x}^{\delta-\gamma_{ij}}S(g_i,g_j) \tag{9}$$

and $\mathrm{mdeg}(\boldsymbol{x}^{\delta-\gamma_{ij}}S(g_i,g_j)) \prec \delta$ for all $1 \le i < j \le s$.

PROOF. By multiplying each c_i with $\mathrm{lc}(g_i)$ if necessary, we may assume that $\mathrm{lc}(g_i) = 1$ and $\mathrm{lt}(g_i) = \mathrm{lm}(g_i) = \boldsymbol{x}^{\mathrm{mdeg}(g_i)}$ for all i. Let $1 \le i < j \le s$. The monomial $\boldsymbol{x}^{\delta} = \boldsymbol{x}^{\alpha_i}\mathrm{lm}(g_i) = \boldsymbol{x}^{\alpha_j}\mathrm{lm}(g_j)$ is a common multiple of $\mathrm{lm}(g_i)$ and $\mathrm{lm}(g_j)$, so that $\boldsymbol{x}^{\gamma_{ij}} \mid \boldsymbol{x}^{\delta}$. Now

$$S(g_i,g_j) = \frac{\boldsymbol{x}^{\gamma_{ij}}}{\mathrm{lt}(g_i)}g_i - \frac{\boldsymbol{x}^{\gamma_{ij}}}{\mathrm{lt}(g_j)}g_j,$$

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

and hence $\operatorname{mdeg}(S(g_i,g_j)) \preccurlyeq \gamma_{ij}$, by Lemma 21.8. Since the leading terms in (7) cancel, we have

$$\operatorname{mdeg}(\boldsymbol{x}^{\delta-\gamma_{ij}}S(g_i,g_j)) = \delta-\gamma_{ij}+\operatorname{mdeg}(S(g_i,g_j)) \prec \delta-\gamma_{ij}+\gamma_{ij}=\delta.$$

We prove (9) by induction on s. If $s=1$, then no cancellation can occur, and the claim is vacuously true. If $s\geq 2$, then

$$\begin{aligned} g &= f - c_1\boldsymbol{x}^{\delta-\gamma_{12}}S(g_1,g_2)\\ &= c_1\boldsymbol{x}^{\alpha_1}g_1 + c_2\boldsymbol{x}^{\alpha_2}g_2 + \sum_{3\leq i\leq s} c_i\boldsymbol{x}^{\alpha_i}g_i - c_1\boldsymbol{x}^{\delta-\gamma_{12}}\left(\frac{\boldsymbol{x}^{\gamma_{12}}}{\operatorname{lt}(g_1)}g_1 - \frac{\boldsymbol{x}^{\gamma_{12}}}{\operatorname{lt}(g_2)}g_2\right)\\ &= c_1(\boldsymbol{x}^{\alpha_1}-\boldsymbol{x}^{\delta-\operatorname{mdeg}(g_1)})g_1 + (c_2\boldsymbol{x}^{\alpha_2}+c_1\boldsymbol{x}^{\delta-\operatorname{mdeg}(g_2)})g_2 + \sum_{3\leq i\leq s} c_i\boldsymbol{x}^{\alpha_i}g_i\\ &= (c_1+c_2)\boldsymbol{x}^{\alpha_2}g_2 + \sum_{3\leq i\leq s} c_i\boldsymbol{x}^{\alpha_i}g_i, \end{aligned}$$

where we use that $\alpha_1+\operatorname{mdeg}(g_1)=\delta=\alpha_2+\operatorname{mdeg}(g_2)$ in the last equation. Now Lemma 21.8 says that $\operatorname{mdeg}(g)\preccurlyeq \max_{\prec}\{\operatorname{mdeg}(f),\operatorname{mdeg}(x^{\delta-\gamma_{12}}S(g_1,g_2))\}\prec\delta$, and hence g is again of the form (8), with a sum of length $s-1$ if $c_1+c_2\neq 0$ and of length $s-2$ otherwise. The induction hypothesis implies that

$$g=\sum_{2\leq i<j\leq s} c_{ij}\boldsymbol{x}^{\delta-\gamma_{ij}}S(g_i,g_j),$$

with some $c_{ij}\in F$ for $2\leq i<j\leq s$; we have $g=0$ if $s=2$. If we let $c_{12}=c_1$ and $c_{1j}=0$ for $3\leq j\leq s$, then

$$f=g+c_1\boldsymbol{x}^{\delta-\gamma_{12}}S(g_1,g_2)=\sum_{1\leq i<j\leq s} c_{ij}\boldsymbol{x}^{\delta-\gamma_{ij}}S(g_i,g_j).\ \square$$

The following theorem gives us an easy test for Gröbner bases.

THEOREM 21.31.
A finite set $G=\{g_1,\dots,g_s\}\subseteq R$ is a Gröbner basis of the ideal $\langle G\rangle$ if and only if

$$S(g_i,g_j)\ \operatorname{rem}\ (g_1,\dots,g_s)=0 \text{ for } 1\leq i<j\leq s. \tag{10}$$

PROOF. "$\Longrightarrow$" follows from Theorem 21.28, since $S(g_i,g_j)\in I=\langle G\rangle$ for all i,j. For the reverse direction, we let $f\in I\setminus\{0\}$, and have to show that $\operatorname{lt}(f)\in\langle\operatorname{lt}(G)\rangle$. We write

$$f=\sum_{1\leq i\leq s} q_ig_i,\quad \delta=\max\nolimits_{\prec}\{\operatorname{mdeg}(q_ig_i)\colon 1\leq i\leq s\}, \tag{11}$$

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

with all $q_i \in R$. Then $\mathrm{mdeg}(f) \preccurlyeq \delta$. If strict inequality holds, then some cancellation of leading terms occurs in (11), and

$$f^* = \sum_{\substack{1\le i\le s \\ \mathrm{mdeg}(q_i g_i)=\delta}} \mathrm{lt}(q_i) g_i$$

is of the form (8). We can write f^* as an F-linear combination of polynomials of the form $\boldsymbol{x}^{\alpha_{ij}} S(g_i, g_j)$, with $\alpha_{ij} \in \mathbb{N}^n$ such that $\alpha_{ij} + \mathrm{mdeg}(S(g_i, g_j)) \prec \delta$ for all i, j, by Lemma 21.30, and then divide by $(g_1, \ldots, g_s)$ with remainder. Since the remainder of $S(g_i, g_j)$ on division by $(g_1, \ldots, g_s)$ is zero, we get $q_i^* \in R$ for $1 \le i \le s$ with $f^* = \sum_{1\le i\le s} q_i^* g_i$, and Theorem 21.12 (ii) implies that $\max_{\prec}\{\mathrm{mdeg}(q_i^* g_i) : 1 \le i \le s\} \prec \delta$. Now both $f - f^*$ and f^* have a representation of the form (11) with a smaller value of δ, and so has f.

Continuing this replacement process if necessary (it terminates because of the well-order property), we find $q_i \in R$ for $1 \le i \le s$ with (11) and $\mathrm{mdeg}(f) = \delta$, and hence $\mathrm{mdeg}(f) = \mathrm{mdeg}(q_i g_i)$ for at least one index i. Then

$$\mathrm{lt}(f) = \sum_{\substack{1\le i\le s \\ \mathrm{mdeg}(q_i g_i)=\delta}} \mathrm{lt}(q_i) \cdot \mathrm{lt}(g_i) \in \langle \mathrm{lt}(G) \rangle. \ \square$$

EXAMPLE 21.32. The **twisted cubic** $C = V(G)$ is the curve in F^3 given by $G = \{y - x^2, z - x^3\}$, so that $C = \{(a, a^2, a^3) : a \in F\} \subseteq F^3$. In $\mathbb{R}^3$, C is the intersection of the two cylindrical surfaces $V(y - x^2)$ and $V(z - x^3)$, as illustrated in Figure 21.5. G is a Gröbner basis for $I = \langle G \rangle$ with respect to the lexicographic order and $y \succ z \succ x$, by Theorem 21.31, since

$$\begin{aligned} S(y - x^2, z - x^3) &= z(y - x^2) - y(z - x^3) = yx^3 - zx^2 \\ &= x^3(y - x^2) + (-x^2)(z - x^3) + 0. \ \diamond \end{aligned}$$

21.5. Buchberger's algorithm

We now describe a method for computing a Gröbner basis, and start by looking at Example 21.21. The basic idea is that, whenever (10) is violated, we add the offending S-polynomial to our basis. Let $\prec = \prec_{\mathrm{grlex}}$ with $y \prec x$, $f_1 = x^3 - 2xy$, and $f_2 = x^2 y - 2y^2 + x \in \mathbb{Q}[x, y]$. $G = \{f_1, f_2\}$ is not a Gröbner basis since $S(f_1, f_2) = -x^2$ and $\mathrm{lt}(S(f_1, f_2)) = -x^2 \notin \langle x^3, x^2 y \rangle = \langle \mathrm{lt}(G) \rangle$. Now we include $f_3 = S(f_1, f_2)$ rem $(f_1, f_2) = -x^2$ in our basis, and $S(f_1, f_2)$ rem $(f_1, f_2, f_3) = 0$.

Next,

$$S(f_1, f_3) = 1 \cdot f_1 - (-x) \cdot f_3 = -2xy,$$

$$S(f_1, f_3) \text{ rem } (f_1, f_2, f_3) = -2xy = f_4,$$

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

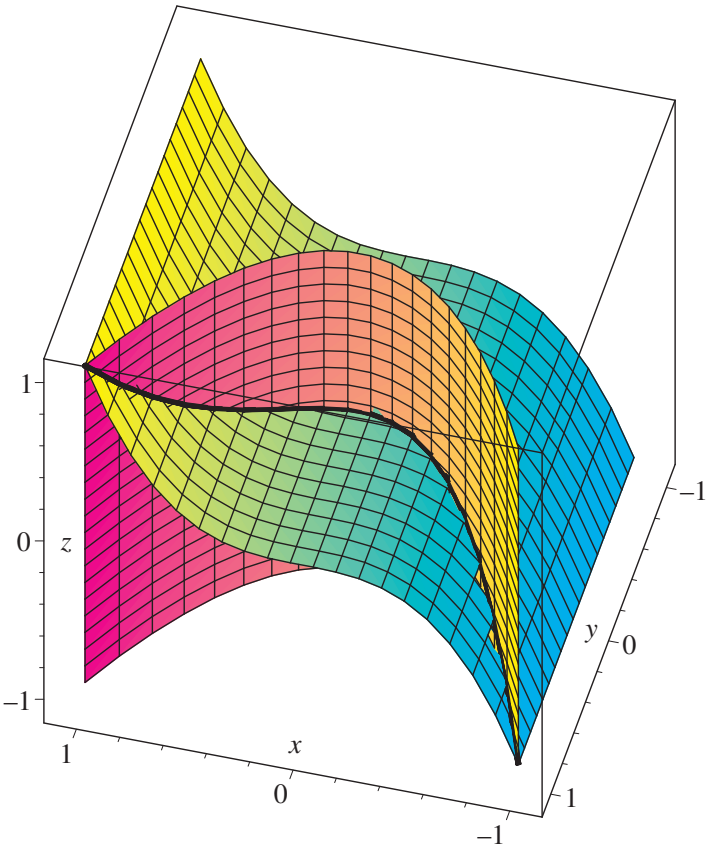


FIGURE 21.5: The twisted cubic (black) in $\mathbb{R}^3$.

which is adjoined to our basis, so that $S(f_1, f_3)$ rem $(f_1, \ldots, f_4) = 0$. Now

$$S(f_1, f_4) = y \cdot f_1 - (-\frac{1}{2}x^2) \cdot f_4 = -2xy^2 = y \cdot f_4,$$

so that $S(f_1, f_4)$ rem $(f_1, \ldots, f_4) = 0$, and

$$S(f_2, f_3) = 1 \cdot f_2 - (-y) f_3 = -2y^2 + x.$$

After adjoining $f_5 = S(f_2, f_3)$ rem $(f_1, \ldots, f_4) = -2y^2 + x$, we check that

$$S(f_i, f_j) \text{ rem } (f_1, \ldots, f_5) = 0 \text{ for } 1 \leq i < j \leq 5,$$

and $\{f_1, \ldots, f_5\}$ is a Gröbner basis.

We now present a simplified version of Buchberger's (1965) algorithm.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

ALGORITHM 21.33 Gröbner basis computation.
Input: $f_1,\dots,f_s \in R = F[x_1,\dots,x_n]$, and a monomial order $\prec$.
Output: A Gröbner basis $G \subseteq R$ for the ideal $I = \langle f_1,\dots,f_s\rangle$ with respect to $\prec$, with $f_1,\dots,f_s \in G$.

1. $G \longleftarrow \{f_1,\dots,f_s\}$
2. **repeat**
3. $\quad \mathcal{S} \longleftarrow \emptyset$
   order the elements of G somehow as $g_1,\dots,g_t$
   for $i = 1,\dots,t-1$ and $j = i+1,\dots,t$ **do**
4. $\quad\quad r \longleftarrow S(g_i,g_j)$ rem $(g_1,\dots,g_t)$
   if $r \neq 0$ **then** $\mathcal{S} \longleftarrow \mathcal{S} \cup \{r\}$
5. $\quad$ **if** $\mathcal{S} = \emptyset$ **then return** G **else** $G \longleftarrow G \cup \mathcal{S}$

THEOREM 21.34.
Algorithm 21.33 works correctly as specified.

PROOF. First we show the correctness assuming that the procedure terminates. At any stage of the algorithm, the set G in step 2 is a basis of I and $f_1,\dots,f_s \in G$, since this is true initially and only elements of I, namely the remainders of S-polynomials of $g_i,g_j \in I$ on division by elements of I, are added to G during the algorithm. If the algorithm terminates, the remainders of all the S-polynomials on division by G are zero, and G is a Gröbner basis by Theorem 21.31.

It remains to show that the algorithm terminates. If G and G^* correspond to successive passes through step 2, then $G^* \supseteq G$ and $\langle \mathrm{lt}(G^*)\rangle \supseteq \langle \mathrm{lt}(G)\rangle$. Hence the ideals $\langle \mathrm{lt}(G)\rangle$ in successive passes through step 2 form an ascending chain, which stabilizes by the ascending chain condition of Corollary 21.24. Thus, after a finite number of steps we have $\langle \mathrm{lt}(G^*)\rangle = \langle \mathrm{lt}(G)\rangle$. We claim that then $G = G^*$. So let $g,h \in G$, and $r = S(g,h)$ rem G. Then $r \in G^*$ and either $r = 0$ or $\mathrm{lt}(r) \in \langle \mathrm{lt}(G^*)\rangle = \langle \mathrm{lt}(G)\rangle$, and from the definition of the remainder we conclude that $r = 0$. □

COROLLARY 21.35.
The following problems are solvable using Gröbner bases:

(i) ideal membership testing via division with remainder,

(ii) ideal containment testing,

(iii) ideal equality testing.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

In general, the Gröbner basis computed by Buchberger's algorithm is neither minimal nor unique. One can, however, manipulate it so that both properties hold.

LEMMA 21.36. *If G is a Gröbner basis of $I \subseteq R$, $g \in G$, and $\mathrm{lt}(g) \in \langle \mathrm{lt}(G \setminus \{g\}) \rangle$, then $G \setminus \{g\}$ is a Gröbner basis of I.*

PROOF. Exercise 21.18. □

DEFINITION 21.37. *A subset $G \subseteq R$ is a* ***minimal*** *Gröbner basis for $I = \langle G \rangle$ if it is a Gröbner basis for I and for all $g \in G$*

(i) $\mathrm{lc}(g) = 1$,

(ii) $\mathrm{lt}(g) \notin \langle \mathrm{lt}(G \setminus \{g\}) \rangle$.

An element g of a Gröbner basis G is ***reduced with respect to G*** *if no monomial of g is in $\langle \mathrm{lt}(G \setminus \{g\}) \rangle$. A minimal Gröbner basis G for $I \subseteq R$ is* ***reduced*** *if all its elements are reduced with respect to G.*

THEOREM 21.38.
Every ideal has a unique reduced Gröbner basis.

PROOF. We first show the existence. Repeatedly applying Lemma 21.36 if necessary, we may start with a minimal Gröbner basis $G = \{g_1, \ldots, g_s\}$. For $1 \leq i \leq s$, we then set

$$h_i = g_i \text{ rem } \{h_1, \ldots, h_{i-1}, g_{i+1}, \ldots, g_s\}.$$

Induction on i proves that $\mathrm{lt}(g_j) = \mathrm{lt}(h_j)$ and h_j is reduced with respect to $G_i = \{h_1, \ldots, h_i, g_{i+1}, \ldots, g_s\}$ for $0 \leq j \leq i \leq s$, and finally $G_s = \{h_1, \ldots, h_s\}$ is a reduced Gröbner basis.

Now suppose that G and G^* are reduced Gröbner bases of I. We claim that $\mathrm{lt}(G) = \mathrm{lt}(G^*)$. For $g \in \mathrm{lt}(G) \subseteq \langle \mathrm{lt}(G) \rangle = \mathrm{lt}(I) = \langle \mathrm{lt}(G^*) \rangle$, there is some $g^* \in G^*$ such that $\mathrm{lt}(g^*) \mid \mathrm{lt}(g)$, by Lemma 21.15. By a symmetric argument, there exists a $g^{**} \in G$ such that $\mathrm{lt}(g^{**}) \mid \mathrm{lt}(g^*)$. Since G is minimal, we have $\mathrm{lt}(g) = \mathrm{lt}(g^*) = \mathrm{lt}(g^{**}) \in \mathrm{lt}(G^*)$, and $\mathrm{lt}(G) \subseteq \mathrm{lt}(G^*)$. Similarly, $\mathrm{lt}(G^*) \subseteq \mathrm{lt}(G)$, which proves the claim.

For a given $g \in G$, let $g^* \in G^*$ be such that $\mathrm{lt}(g) = \mathrm{lt}(g^*)$. Both G and G^* are reduced, and hence no monomial in $g - g^* \in I$ is divisible by any element of $\mathrm{lt}(G) = \mathrm{lt}(G^*)$. Thus $g - g^* = g - g^* \text{ rem } G = 0$ since $g - g^* \in I$, whence $g \in G^*$, $G \subseteq G^*$, and by a symmetric argument, also $G^* \subseteq G$. □

At the beginning of this section, we saw how several polynomials may have to be added to form a Gröbner basis. How many? In Section 21.7, we will learn

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

the rather devastating answer: sometimes doubly exponentially many, and their degrees may be doubly exponentially large (in the number of variables). It is not easy to say how many steps Buchberger's algorithm takes, but for such huge outputs it uses at least exponential space.

Both Gaussian elimination and Euclid's algorithm for gcds in $F[x]$ are special cases of Buchberger's algorithm (see Exercise 21.24 for the former).

21.6. Geometric applications

Algebra provides a powerful language which can express a wide range of problems. Then a general–purpose solution strategy like Gröbner bases is available for the solution of these problems. Of course, this approach has its limitations, for example, with regards to efficiency. We now give two sample applications from geometry. In Chapter 24, we will exhibit two computer science questions whose geometric formulation is not obvious (but easy to produce, once we set our mind on it), and a larger example from chemistry.

Automatic geometric theorem proving. Geometric theorems can often be formulated in terms of polynomial equations. We have seen this for the theorem about the center of gravity of a triangle in Example 21.2, where the hypotheses yielded two polynomials $f_1, f_2 \in \mathbb{R}[u,v,x,y]$, the conclusions three further polynomials $g_1, g_2, g_3 \in \mathbb{R}[u,v,x,y]$, and the theorem is equivalent to "$f_1 = f_2 = 0 \Longrightarrow g_1 = g_2 = g_3 = 0$". In general, we obtain a set of hypothesis polynomials $f_1, \ldots, f_s$ in $R = \mathbb{R}[x_1, \ldots, x_n]$ and one (or several) conclusion polynomials $g \in R$, and the theorem is true if and only if $V(f_1, \ldots, f_s) \subseteq V(g)$. In particular, we can prove the theorem by showing that $g \in \langle f_1, \ldots, f_s \rangle$.

In Example 21.2, we have already done this for $g = g_1$, so let us try Gröbner bases on g_2 and g_3. We start by computing a Gröbner basis for the ideal $I = \langle f_1, f_2 \rangle$ with respect to $\prec = \prec_{\text{lex}}$ and $u \succ v \succ x \succ y$. We have

$$f_1 = uy - vx - v, \quad f_2 = uy - vx + 2v - y$$

with respect to this monomial order,

$$S(f_1, f_2) = f_1 - f_2 = -3v + y = -g_3, \quad -g_3 \text{ rem } (f_1, f_2) = -g_3.$$

After adding $f_3 = -g_3$ to our basis, it turns out that

$$S(f_1, f_3) \text{ rem } (f_1, f_2, f_3) = \frac{1}{3}uy^2 - v^2x - v^2 \text{ rem } (f_1, f_2, f_3) = 0,$$
$$S(f_2, f_3) \text{ rem } (f_1, f_2, f_3) = \frac{1}{3}uy^2 - v^2x + 2v^2 - vy \text{ rem } (f_1, f_2, f_3) = 0,$$

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

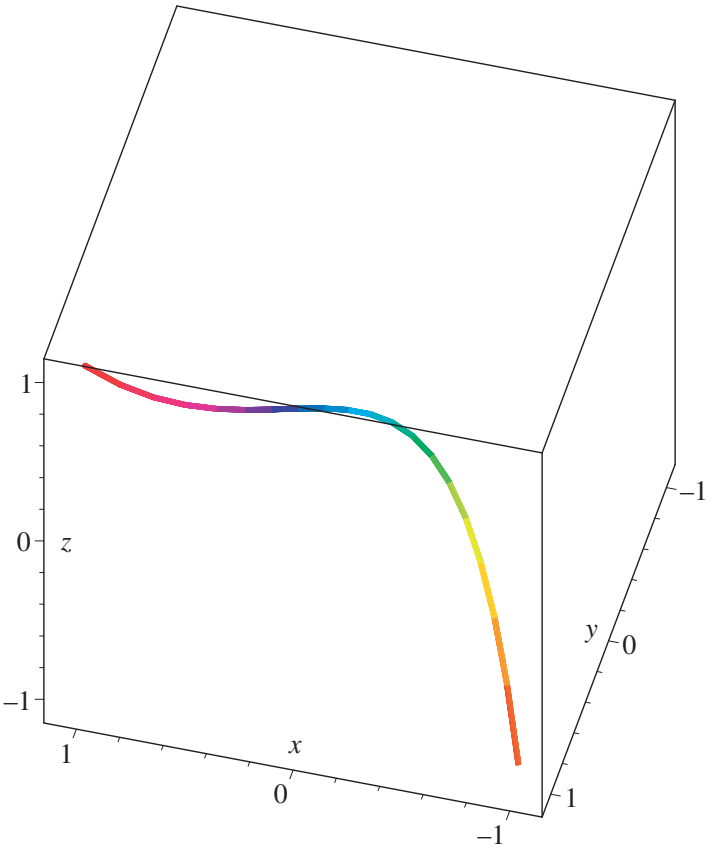


FIGURE 21.6: The twisted cubic.

and $\{f_1, f_2, f_3\}$ is a Gröbner basis. By Lemma 21.36, we may discard f_2, and using the process from the proof of Theorem 21.38 for reducing, we obtain

$$f_4 = f_1 \text{ rem } f_3 = uy - \frac{1}{3}xy - \frac{1}{3}y, \quad f_3 \text{ rem } f_4 = f_3,$$

and $G = \{f_4, f_3\} = \{uy - \frac{1}{3}xy - \frac{1}{3}y, v - \frac{1}{3}y\}$ is the unique reduced Gröbner basis of I with respect to $\prec$.

We know already that $g_3 = -f_3 \in I$. Division with remainder yields g_2 rem $G = g_2 \neq 0$, and Theorem 21.28 implies that $g_2 \notin I$. On the other hand, $f_4 = \frac{1}{3}yg_2$ belongs to I, and hence the theorem about the center of gravity is true if $y \neq 0$. (See Notes 21.6 for the degenerate case $y = 0$.)

Implicitization. Let $f_1, \ldots, f_n \in F[t_1, \ldots, t_m]$, and suppose that the affine algebraic variety $V \subseteq F^n$ is given in parametrized form by

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

$$\begin{aligned} x_1 &= f_1(t_1,\dots,t_m),\\ &\vdots\\ x_n &= f_n(t_1,\dots,t_m), \end{aligned}$$

so that V is "explicitly" described as $V = \{a \in F^n : \exists b \in F^m\ a = (f_1(b),\dots,f_n(b))\}$. The task is now to find polynomials $g_1,\dots,g_s \in F[x_1,\dots,x_n]$ such that V has the "implicit" representation $V = V(I)$, where $I = \langle g_1,\dots,g_s\rangle$. (More precisely, $V(I)$ will equal the "closure" of V.)

EXAMPLE 21.32 (continued). The twisted cubic C from Example 21.32 can be parametrized by

$$x = t, \quad y = t^2, \quad z = t^3.$$

An implicitization for C is $g_1 = y - x^2, g_2 = z - x^3$. The curve is illustrated in Figures 21.5 and 21.6 on pages 609 and 613, respectively. The latter corresponds to the explicit representation of C (the plot was generated by letting the parameter t run through the interval $[-1,1]$), while the former depicts the implicit form as the intersection of the two surfaces defined by g_1 and g_2; this picture is somehow more informative. $\diamond$

EXAMPLE 21.39. Let $V \subseteq F^3$ be parametrized by $x = t^2$, $y = t^3$, $z = t^4$. Then an implicitization for V is $g_1 = z - x^2, g_2 = y^2 - x^3$. $\diamond$

To solve the implicitization problem, we consider $J = \langle x_1 - f_1,\dots,x_n - f_n\rangle \subseteq F[t_1,\dots,t_m,x_1,\dots,x_n]$, order the variables so that $t_1 \succ \cdots \succ t_m \succ x_1 \succ \cdots \succ x_n$, and compute a Gröbner basis G for J with respect to $\prec = \prec_{\text{lex}}$. Then some $g \in G$ will depend only on $x_1,\dots,x_n$. These are candidates for the g_i's.

In Examples 21.32 and 21.39, the reduced Gröbner bases for J with respect to $t \succ z \succ y \succ x$ are $\{t - x, z - x^3, y - x^2\}$ and $\{t^2 - x, ty - x^2, tx - y, z - x^2, y^2 - x^3\}$, respectively.

This approach always works over an infinite field, in the sense that the variety in F^n defined by the polynomials in $G \cap F[x_1,\dots,x_n]$ is the smallest variety containing the image of the parametrization (see Cox, Little & O'Shea 1997, §3.3).

Solving systems of polynomial equations. This is the most natural application of Gröbner bases, and they were originally invented for this. In general, we have polynomials $f_1,\dots,f_m \in F[x_1,\dots,x_n]$ and want to answer questions about the corresponding affine variety $V = \{a \in F^n : f_1(a) = \cdots = f_m(a) = 0\}$. Examples of such questions are: Is $V \neq \emptyset$? If so, please find a point on V. Given another polynomial $g \in F[x_1,\dots,x_n]$, is $g(a) = 0$ for all $a \in V$?

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

One of the simplest situations occurs when V is **zero-dimensional**, so that it consists only of finitely many isolated points. An example is the intersection of two plane curves. We saw in Section 6.8 how to use resultants for this problem: one first finds a polynomial $r \in F[x]$ in one variable which the x–coordinates of all intersection points satisfy, computes all roots of r, and then has to solve, for each root, the original bivariate equations (or their gcd). As mentioned, this resultant approach has a (nontrivial) generalization to higher dimensions.

Gröbner bases provide an alternative approach. For a zero–dimensional problem, the Gröbner basis with respect to lexicographic order $\prec = \prec_{lex}$ and $x_1 \prec \cdots \prec x_n$ will often include some polynomials only containing x_1, some only with x_1, x_2, and so on. These can then be solved by back substitution: first for x_1, then for x_2, and so on.

Application areas such as robot kinematics and motion planning provide many such interesting geometric problems. We now look at a simple example: the intersection of two plane curves. A more elaborate application in chemistry is discussed in Section 24.4.

We let $f = (y^2+6)(x-1) - y(x^2+1)$ and $g = (x^2+6)(y-1) - x(y^2+1) = f(y,x)$ in $\mathbb{C}[x,y]$, as in Example 21.3 (ii), and use Gröbner bases to determine all intersection points of the two plane curves $V(f)$ and $V(g)$, or equivalently, all common roots of the system of polynomial equations $f = g = 0$. We start by computing the reduced Gröbner basis G of $I = \langle f, g \rangle$ with respect to the lexicographic order $\prec = \prec_{\text{lex}}$ and $x \prec y$. We find that G consists of f, g, and the three polynomials

$$h = -f - g = x^2 - 5x + y^2 - 5y + 12, \quad p = y^2x - 5yx + 6x + y^3 - 6y^2 + 11y - 6,$$

$$q = y^4 - 6y^3 + 15y^2 - 26y + 24.$$

The last polynomial contains only the variable y, and we determine all its roots symbolically by factoring it over $\mathbb{Q}$. Using one of the algorithms from Part III, we obtain $q = (y-2)(y-3)(y^2 - y + 4)$, and hence

$$V(q) = \left\{2, 3, \frac{1 \pm \sqrt{15}i}{2}\right\}.$$

It remains to determine which of these partial solutions extend to a solution in $V(h,p,q) = V(I)$. Substituting $y = 2$ in h and p, we find that $p(x,2)$ vanishes identically, and from the solutions of the univariate equation

$$h(x,2) = x^2 - 5x + 6 = (x-2)(x-3) = 0,$$

for x, we find the two common zeroes $(3,2)$ and $(3,3)$ of f and g in $\mathbb{C}^2$. The other four intersection points from (2) can be obtained in a similar fashion, by substituting $y = 2$ or $y = (1 \pm \sqrt{15}i)/2$ in $h = p = 0$ and solving for x.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

21.7. The complexity of computing Gröbner bases

For over thirty years after its publication in 1965, the cost of computing Gröbner bases remained a mystery. In this section, we briefly recount its solution by Mayr and his collaborators, and some remaining open questions. The limited space of this text prevents us from giving any details.

Experiments with implementations almost invariably ran into the dreaded phenomenon of **intermediate expression swell**. Starting out with a few polynomials of low degree and with small coefficients, the systems produce huge numbers of polynomials with very large degrees and enormous coefficients. In fact, the solution tells us that this is so by necessity, at least in the worst case. This should be contrasted with the situation for Euclid's algorithm in $\mathbb{Q}[x]$, where also intermediate expression swell was empirically observed, but then kept under control, using subresultants and clever algorithmic variations (Chapter 6).

We need some notions from complexity theory, as explained in Section 25.8. We have "feasible" classes like $\mathcal{P}$ and $\mathcal{BPP}$, the infeasible $\mathcal{NP}$-complete problems (assuming $\mathcal{P} \neq \mathcal{NP}$), and much "higher up" the class $\mathcal{EXPSPACE}$ of problems that can be solved using exponential space. The $\mathcal{EXPSPACE}$-complete problems are exceedingly difficult; they typically use doubly exponential time $2^{2^{O(n)}}$, for some inputs of size n at least.

We write IM for the polynomial ideal membership problem over $\mathbb{Q}$. Already Mayr & Meyer (1982) had shown that IM is $\mathcal{EXPSPACE}$-hard for general ideals. Mayr (1989, 1992) proved that IM is in $\mathcal{EXPSPACE}$, and thus $\mathcal{EXPSPACE}$-complete. Since Theorem 21.28 provides a reduction from IM to computing a Gröbner basis, the (decision version of the) latter problem is also $\mathcal{EXPSPACE}$-hard. The question was finally settled by Mayr (1989) and Kühnle & Mayr (1996), who gave an exponential space algorithm for computing a (reduced) Gröbner basis.

THEOREM 21.40.
The problem of finding a reduced Gröbner basis is $\mathcal{EXPSPACE}$-complete.

The same results hold for **binomial ideals** (generated by binomials $\boldsymbol{x}^{\alpha} - \boldsymbol{x}^{\beta}$), and for some other types of fields besides $\mathbb{Q}$. Bürgisser (1998) proved that over any infinite field, the ideal membership problem requires exponential parallel time. For homogeneous ideals (generated by polynomials f each of whose terms has total degree $\deg f$), Mayr (1995) proved that IM is $\mathcal{PSPACE}$-complete, while computing a Gröbner basis is $\mathcal{EXPSPACE}$-complete.

The upper bounds mentioned make use of estimates of certain degrees. Ideal membership was first proven to be decidable by Hilbert (1890). Hermann (1926) gave a constructive method to represent any $f \in \langle f_1, \ldots, f_s \rangle$ as a linear combination

$$f = \sum_{1 \leq i \leq s} q_i f_i. \tag{12}$$

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

She gave a doubly exponential bound on the degrees of these q_i's; see also Mayr & Meyer (1982). Mayr & Ritscher (2010) showed that the degrees of the polynomials in a reduced Gröbner basis for an ideal in $F[x_1, \ldots, x_n]$ of dimension r are at most

$$2\left(\frac{d^{n-r}}{2} + d\right)^{2^r}, \tag{13}$$

when $\deg f_i \leq d$ for all i. This bound does not depend on the number of polynomials defining the ideal nor on their coefficients, depends polynomially on the degree, exponentially on the codimension and doubly exponentially on the dimension. They also show a lower bound of a similar form.

If an ideal is such that the degrees in its reduced Gröbner basis come close to the doubly exponential bound (13), one might think that the output cannot even be written down in exponential space, and similarly for the certificate (12) of ideal membership. However, the model of space-bounded computations is such that one may take doubly exponential time to write a result of doubly exponential length on a special output tape, all the while using only singly exponential work space (Section 25.8).

Hilbert's famous **Nullstellensatz** says the following. If F is algebraically closed (say, $F = \mathbb{C}$), $f, f_1, \ldots, f_s \in F[x_1, \ldots, x_n]$, and $f(a) = 0$ for all $a \in F^n$ with $f_1(a) = \cdots = f_s(a) = 0$, then there exists $e \in \mathbb{N}$ with $f^e \in \langle f_1, \ldots, f_s \rangle$. In particular, the variety $V(f_1, \ldots, f_s)$ is empty if and only if $1 \in \langle f_1, \ldots, f_s \rangle$. If this is the case, then 1 will appear in any Gröbner basis of $f_1, \ldots, f_s$. This provides a test of whether $V(f_1, \ldots, f_s)$ is empty. For this particular instance of IM better results are available: one can always choose e and the degrees in (12) to be simply exponential, and this implies that the problem is in $\mathcal{PSPACE}$.

The worst-case cost of Buchberger's algorithm is still unknown today, but the important result of Theorem 21.40 completely settles the question of the "best" worst-case cost of any algorithm for Gröbner bases, and provides a lower bound for Buchberger's. It gives rise to the pessimistic view that these methods for polynomial ideals are not useful in practice, except for rather small cases.

However, it is not the full story. The inputs used for the lower bound are more combinatorial than geometric in nature, while most of the problems that people try to solve derive from geometric tasks. The algorithm of Kühnle & Mayr (1996) uses essentially the same time for all polynomials of a given degree and number of variables and thus is uniformly impractical, while one might hope that "natural" geometric problems are easier to solve than "combinatorial" ones.

Notes. **21.1.** The papers in Eisenbud & Robbiano (1993) present the state of the art at that time. A good reference for this chapter is Cox, Little & O'Shea (1997), which we have followed closely in our exposition. Cox, Little & O'Shea (1998) discuss more advanced topics.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

21.3. With the passage of time, the proof of Hilbert's basis theorem has become quite simple. But his 1890 paper was a milestone, solving this long–standing open question, and furthermore introducing the Hilbert function of an ideal and showing that invariant rings are finitely generated.

21.5. Buchberger (1965,1970,1976,1985,1987) explains his Gröbner basis method, gives many references, and puts it in the context of more general questions about grammars and term writing systems. Two further important contributions of Bruno Buchberger are the founding of the Journal of Symbolic Computation in 1985 and of the Research Institute for Symbolic Computation (RISC) in Linz, Austria.

21.6. Buchberger & Winkler (1998) contains a variety of tutorials on applications of Gröbner bases. With slight modifications, the approach described in the text works for a rich class of geometric theorems (see Cox, Little & O'Shea 1997, §6.4, and Wu 1994).

In Section 21.6, we found that $g_2 \notin I$ and $yg_2 \in I$. Thus we may conclude that $g_2(x,y) = 0$ if $(x,y) \in V(I)$ and $y \neq 0$. This can be phrased as an ideal membership property via *Rabinowitsch's trick* (1930) of adding $1 - yz$ to I, where z is a new indeterminate. This ensures that the value of y is nonzero, and $g_2 = 3z \cdot f_4 + g_2 \cdot (1 - yz) \in \langle f_3, f_4, 1 - yz \rangle$.

The reverse question of transforming an implicit representation of a variety into an explicit one has in general no solution; it can be solved only for the "rational varieties" of genus zero. However, it can be solved "near a smooth point" in general, if we allow appropriate power series for the f_i's, as in the implicit function theorem of calculus. The state of the art about algorithms on parametric varieties is presented in the twelve articles of a Special Issue of the *Journal of Symbolic Computation* (Hoffman, Sendra & Winkler 1997).

It can be shown that for zero-dimensional ideals $I \in F[x_1, \ldots, x_n]$, so that $V(I)$ is finite, a Gröbner basis with respect to lexicographic order $x_1 \prec x_2 \prec \cdots \prec x_n$ always contains a "triangular" subset $g_1, \ldots, g_n$, such that $g_i \in F[x_1, \ldots, x_i]$ and $\mathrm{lm}(g_i)$ is a power of x_i, for $1 \leq i \leq n$ (see, for example, Becker & Weispfenning (1993), Theorem 6.54, or Cox, Little & O'Shea (1997), §3.1 and 3.2).

21.7. Yap (1991) gave an improvement of Mayr & Meyer's result. Brownawell (1987) and Kollár (1988) proved that one can always choose the Nullstellensatz exponent e to be simply exponential; see also Amoroso (1989). Caniglia, Galligo & Heintz (1989), Lakshman (1990), and Berenstein & Yger (1990) showed that in some important cases the q_i's in (12) have only singly exponential degree: for zero-dimensional varieties, and for complete intersections.

Giusti (1984), Möller & Mora (1984), Bayer & Stillman (1988), and Dubé (1990) proved upper bounds for the elements of a reduced Gröbner basis. Huynh (1986) showed a lower bound on the number and degrees of polynomials in Gröbner bases. Mayr (1997) gives a survey of complexity results, more references, and discusses applications such as those in Section 24.2.

Bayer & Stillman (1988) considered an invariant m associated to any multivariate ideal, called the *Castelnuovo-Mumford regularity*. This number seems to be fairly small for many natural geometric problems, but is exponential in the number of variables for the combinatorial problems of Mayr & Meyer (1982). Furthermore, Bayer & Stillman prove that, after a generic change of coordinates, the polynomials in a Gröbner basis with respect to the graded reverse lexicographic order $\prec_{\mathrm{grevlex}}$ have degree at most m. This gives rise to a bit of hope that the method might be able to deal successfully with interesting geometric problems, and to a practical recommendation in favor of $\prec_{\mathrm{grevlex}}$.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

Almost all computer algebra systems contain some routines for Gröbner bases; Dave Bayer's system MACAULAY focuses particularly on this problem, and SINGULAR is another powerful package in this area. The research projects POSSO and FRISCO of the European Community have produced substantial software and a library of benchmark problems. Efficient algorithms and software are a highly active area of research. Three topics of particular interest are modular algorithms (Traverso 1988), selection strategies for S-polynomials (Giovini, Mora, Niesi, Robbiano & Traverso 1991), and conversion between Gröbner bases with respect to different orders (Faugère, Gianni, Lazard & Mora 1993).

Finally, we mention that there are other methods of dealing with certain geometric problems, based on cylindrical algebraic decomposition (Collins 1975), elimination theory, also using arithmetic circuits as a data structure (Chistov & Grigor'ev 1984, Caniglia, Galligo & Heintz 1988, Fitchas, Galligo & Morgenstern 1990, Giusti & Heintz 1991), u-resultants (Macaulay 1902, 1916, 1922, Canny 1987), and characteristic sets (Ritt 1950, Wu 1994, Gallo & Mishra 1991).

The important subject of *computational real algebraic geometry* started with Tarski (1948). Major progress was made by Collins (1975), and later by Ben-Or, Kozen & Reif (1986), Fitchas, Galligo & Morgenstern (1987), Grigor'ev (1988), Canny (1988), Renegar (1992a, 1992b, 1992c), and others. See the surveys of Heintz, Recio & Roy (1991) and Renegar (1991) for references and applications; our cyclohexane example in Section 24.4 can be viewed as such an application.

Exercises.

21.1 Let F be a field and x, y indeterminates. Prove that the two ideals $\langle x, y\rangle$ and $\langle \gcd(x, y)\rangle$ in $F[x, y]$ are distinct, and conclude that $F[x, y]$ is not Euclidean. Hint: Exercise 3.17.

21.2 Let F be a field. Prove that the ideals $I = \langle x + xy, y + xy, x^2, y^2\rangle$ and $J = \langle x, y\rangle$ in $F[x, y]$ are identical. Your proof should also work if $\operatorname{char} F = 2$. Hint: It is sufficient to prove that the generators of each ideal belong to the other ideal.

21.3 Prove (3). Hint: Theorem 4.11.

21.4* Besides the usual Cartesian coordinates (u, v) with $u, v \in \mathbb{R}$, we represent the points of the plane by polar coordinates (r, φ) with $r \in \mathbb{R}$ and $0 \le \varphi < 2\pi$. This representation is not unique; for example, when $\varphi < \pi$ then (r, φ) and $(-r, \varphi + \pi)$ represent the same point. We obtain the polar coordinates from the Cartesian ones by the formulas $u = r\cos\varphi$, and $v = r\sin\varphi$. Now consider the curve $C = \{(r, \varphi) : 0 \le \varphi < 2\pi \text{ and } r = \sin 2\varphi\} \subseteq \mathbb{R}^2$, and let $I = \langle (x^2 + y^2)^3 - 4x^2y^2\rangle \subseteq \mathbb{R}[x, y]$.

(i) Create a plot of C.

(ii) Using the addition formulas for sine and cosine, show that $C \subseteq V(I)$.

(iii) Prove that also the reverse inclusion $V(I) \subseteq C$ holds (be careful with the signs).

21.5* Let F be a field and $n \in \mathbb{N}$. For a subset $M \subseteq F^n$, we define the **ideal of** M by

$$\mathbf{I}(M) = \{f \in F[x_1, \ldots, x_n] : f(u) = 0 \text{ for all } u \in M\}.$$

(i) Show that $\mathbf{I}(M)$ is in fact an ideal.

(ii) Prove that $N \subseteq M \Longrightarrow \mathbf{I}(N) \supseteq \mathbf{I}(M)$ and $V(\mathbf{I}(M)) \supseteq M$ for all $M, N \subseteq F^n$.

(iii) Prove that $\mathbf{I}(M)$ is **radical**, which means that $f^m \in I$ implies $f \in I$ for all $m \in \mathbb{N}$ and $f \in F[x_1, \ldots, x_n]$.

(iv) Let $P = (u, v) \in F^2$ be an arbitrary point. Determine polynomials $f_1, f_2 \in F[x, y]$ such that $\mathbf{I}(\{P\}) = \langle f_1, f_2\rangle$. Find a set $M \subseteq F^2$ such that $\mathbf{I}(M) = \langle x, y\rangle$.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

(v) Determine $\mathbf{I}(\varnothing)$ and $\mathbf{I}(F^n)$. Hint: Start with $n = 1$.

(vi) Let $M = \{(u,0) \in \mathbb{R}^2 : 0 \le u \le 1\} \subseteq \mathbb{R}^2$, and find a polynomial $f \in \mathbb{R}[x,y]$ with $\mathbf{I}(M) = \langle f \rangle$.

21.6 Let $f = 2x^4y^2z - 6x^4yz^2 + 4xy^4z^2 - 3xy^2z^4 + x^2y^4z - 5x^2yz^4$ in $\mathbb{Q}[x,y,z]$.

(i) Determine the order of the monomials in f for the three monomial orders $\prec_{\text{lex}}$, $\prec_{\text{grlex}}$, and $\prec_{\text{grevlex}}$, with $x \succ y \succ z$ in all cases.

(ii) For each of the three monomial orders from (i), determine $\operatorname{mdeg}(f)$, $\operatorname{lc}(f)$, $\operatorname{lm}(f)$, and $\operatorname{lt}(f)$.

21.7 Let $<$ be an order on a set S. Prove that $\alpha < \beta$ implies $\beta \not< \alpha$ for all $\alpha, \beta \in S$.

21.8 Show that $0 = (0,\dots,0) \in \mathbb{N}^n$ is the smallest element with respect to a monomial order $\prec$: $0 = \min_{\prec} \mathbb{N}^n$.

21.9$^{\longrightarrow}$ (i) Prove that $\prec_{\text{lex}}$ is a monomial order.

(ii) Let $\prec$ be a total order on R, not necessarily a well–order. Prove that the graded variant of $\prec$ defined by

$$\alpha \prec_{\text{gr}} \beta \iff \sum_{1\le i\le n} \alpha_i < \sum_{1\le i\le n} \beta_i \text{ or } \left(\sum_{1\le i\le n} \alpha_i = \sum_{1\le i\le n} \beta_i \text{ and } \alpha \prec \beta \right)$$

is a monomial order, and conclude that so is $\prec_{\text{grlex}}$.

21.10 How many monomials in variables $x_1,\dots,x_n$ have total degree m?

21.11 Prove Lemma 21.8.

21.12 Prove Theorem 21.12.

21.13 Let $n \in \mathbb{N}$ and $\alpha = (\alpha_1,\dots,\alpha_n) \in \mathbb{N}^n$. Determine the number of elements $\beta = (\beta_1,\dots,\beta_n) \in \mathbb{N}^n$ such that $\beta_i \le \alpha_i$ for $1 \le i \le n$.

21.14 Show that the set B from the proof of Dickson's lemma is the smallest subset of A (with respect to inclusion) such that $\langle \boldsymbol{x}^B \rangle = \langle \boldsymbol{x}^A \rangle$.

21.15 Show that for each $n \in \mathbb{N}$ there exists a monomial ideal $I \subseteq \mathbb{Q}[x,y]$ such that every basis of I has at least n elements.

21.16 Show that G from Example 21.32 is not a Gröbner basis with respect to $\prec_{\text{lex}}$ and $x \succ y \succ z$.

21.17 Let F be a field, $R = F[x_1,\dots,x_n]$, and $f_1,\dots,f_s \in R$. Prove that $\{f_1,\dots,f_s\}$ is a Gröbner basis if $f \operatorname{rem} (f_1,\dots,f_s) = 0$ for all $f \in I = \langle f_1,\dots,f_s \rangle$.

21.18 Prove Lemma 21.36.

21.19 Let G be a Gröbner basis for the ideal $I \subseteq F[x_1,\dots,x_n]$, where F is a field. Prove that $1 \in I$ if and only if G contains a nonzero constant from F. Conclude that $G = \{1\}$ if $1 \in I$ and G is reduced.

21.20 (i) Compute a Gröbner basis for the ideal $I = \langle x^2 + y - 1, xy - x \rangle \subseteq \mathbb{Q}[x,y]$ with respect to $\prec = \prec_{\text{lex}}$ and $x \succ y$, using Buchberger's algorithm. If your basis is not minimal, then use Lemma 21.18 to compute a minimal one.

(ii) Which of the following polynomials belong to I: $f_1 = x^2 + y^2 - y$, $f_2 = 3xy^2 - 4xy + x + 1$?

21.21 Which of the following finite subsets of $\mathbb{Q}[x,y,z]$ are Gröbner bases with respect to $\prec = \prec_{\text{lex}}$? Which are minimal or even reduced?

(i) $\{x+y, y^2-1\}$ for $x \succ y$,

(ii) $\{y+x, y^2-1\}$ for $y \succ x$,

(iii) $\{x^2+y^2-1, xy-1, x+y^3-y\}$ for $x \succ y$,

(iv) $\{xyz-1, x-y, y^2z-1\}$ for $x \succ y \succ z$.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

21.22 Let F be a field and $\{f_1,\ldots,f_s\}$ and $\{g_1,\ldots,g_t\}$ in $R=F[x_1,\ldots,x_n]$ be minimal Gröbner bases of the same ideal $I\subseteq R$, with $f_1\preccurlyeq\cdots\preccurlyeq f_s$ and $g_1\preccurlyeq\cdots\preccurlyeq g_t$. Prove that $s=t$ and $\mathrm{lt}(f_i)=\mathrm{lt}(g_i)$ for all i.

21.23$^{\longrightarrow}$ Compute a Gröbner basis for

$$\langle f_1=x^2y-2yz+1, f_2=xy^2-z^2+2x, f_3=y^2z-x^2+5\rangle\subseteq\mathbb{Q}[x,y,z],$$

using $\prec=\prec_{\mathrm{grlex}}$ with $x\prec y\prec z$. Compare your output to the Gröbner basis that MAPLE computes with a different order.

21.24* Let F be a field, $n\in\mathbb{N}$, and $A=(a_{ij})_{1\le i,j\le n}\in F^{n\times n}$ a square matrix. Moreover, let $G_A=\{\sum_{1\le j\le n}a_{ij}x_j\colon 1\le i\le n\}\subseteq F[x_1,\ldots,x_n]$ be the set of linear polynomials corresponding to the rows of A and $I_A=\langle G_A\rangle$. Then $V(G_A)=V(I_A)$ is equal to $\ker A$, the set of solutions $v\in F^n$ of the linear system $Av=0$. Prove:

(i) $I_{LA}=I_A$ if $L\in F^{n\times n}$ is nonsingular.

(ii) Assume that there exists a nonsingular matrix $L\in F^{n\times n}$ such that

$$U=LA=\begin{pmatrix} I_r & V\\ 0 & 0\end{pmatrix},$$

where r is the rank of A, I_r is the $r\times r$ identity matrix, and $V\in F^{r\times(n-r)}$ (this means that no column exchange is necessary when applying Gaussian elimination to A). Prove that G_U is a reduced Gröbner basis of I_A with respect to any monomial order $\prec$ such that $x_1\succ x_2\succ\cdots\succ x_n$.

(iii) What is the reduced Gröbner basis of I_A if A is nonsingular, with respect to an arbitrary monomial order?

21.25* You are to solve the following nonlinear optimization problem: Determine all maxima and minima of the polynomial $f=x^2y-2xy+y+1$ on the unit circle $S=\{(u,v)\in\mathbb{R}^2: g(u,v)=0\}$, where $g=x^2+y^2-1$. In numerical analysis, such a problem is solved with the aid of **Lagrange multipliers**: if $\nabla f=(f_x,f_y)$ and $\nabla g=(g_x,g_y)$ are the Jacobians of f and g, respectively, where $f_x=\partial f/\partial x$ and f_y, g_x, g_y, are defined analogously, then the equality $\nabla f=\lambda\nabla g$ holds at a local maximum or minimum of f on S for some $\lambda\in\mathbb{R}$.

(i) Set up the system of polynomial equations

$$g=0 \text{ and } \nabla f-z\nabla g=0, \tag{14}$$

where z is another indeterminate.

(ii) Let $I\subseteq\mathbb{Q}[x,y,z]$ be the ideal generated by the equations in (14). Compute a Gröbner basis G of I with respect to $\prec=\prec_{\mathrm{lex}}$ and $z\succ x\succ y$.

(iii) Solve the system of polynomial equations corresponding to G, which is equivalent to (14), by back substitution. Check that all solutions are in fact solutions of (14). Which are the absolute maxima and minima of f on the unit circle?

(iv) Generate a plot of the values of f on S.

21.26 Find an example over $\mathbb{F}_5$ (which is not algebraically closed) where the statement of Hilbert's Nullstellensatz is false.

21.27 Prove Hilbert's Nullstellensatz for univariate polynomials over $\mathbb{C}$.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press

Schönheit, höre ich Sie da fragen; entfliehen nicht die Grazien,
wo Integrale ihre Hälse recken.[1]

Ludwig Boltzmann (1887)

Es dürfte richtig sein, zu sagen, daß die Begriffe Differentialquotient
und Integral, deren Usprung jedenfalls bis auf Archimedes zurückgeht,
dem Wesen der Sache nach durch die Untersuchungen von Kepler,
Descartes, Cavalieri, Fermat und Wallis in die Wissenschaft
eingeführt worden sind. [. . .] Sie hatten noch nicht bemerkt,
daß Differentiation und Integration *inverse* Operationen sind.
Diese kapitale Entdeckung gehört Newton und Leibniz.[2]

Marius Sophus Lie (1895)

Common integration is only the *memory of differentiation*
[. . .] the artifices by which it [integration] is effected, are changes,
not from the unknown to the known, but from the forms
in which memory will not serve us to those in which it will.

Augustus De Morgan (1844)

He who can digest a second or third Fluxion,
a second or third Difference, need not, methinks,
be squeamish about any Point in Divinity.

George Berkeley (1734)

The Integration of Quantities seems to merit farther labour and
research; and no doubt this important and abstruse branch will,
by and by, obtain due consideration,
and we shall have important simplifications of,
and additions to, our already large stock of knowledge.

William Shanks (1853)

[1] Beauty, I hear you ask; do not the Graces flee where integrals stretch forth their necks?

[2] It may be said that the notions of differential quotient and integral, whose origin certainly goes back to Archimedes, were essentially introduced into science by the investigations of Kepler, Descartes, Cavalieri, Fermat, and Wallis. [. . .] They had not yet noticed that differentiation and integration are *inverse* operations; this capital discovery belongs to Newton and Leibniz.

https://doi.org/10.1017/CBO9781139856065.027 Published online by Cambridge University Press